CHAPTER FOUR

KINETICS OF A PARTICLE: WORK AND ENERGY

Learning Objectives

After completing this chapter, students should be able to do the following:

- To develop the principle of work and energy and apply it to solve problems that involve force, velocity, and displacement.

4.1 Work of a Variable Force

If the particle acted upon by the force **F** undergoes a finite displacement along its path from r_1 to r_2 or s_1 to s_2 , Fig. 4.1a,

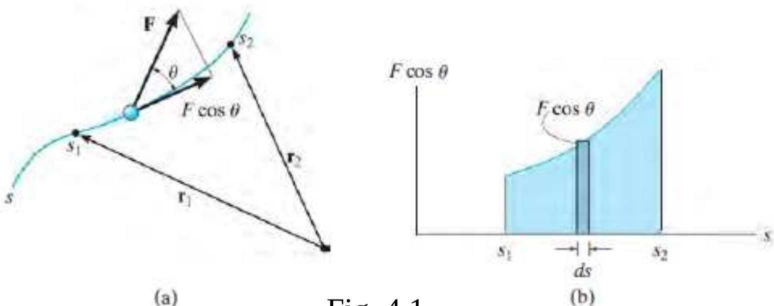


Fig. 4.1

the work of force **F** is determined by integration. Provided **F** and θ can be expressed as a function of position, then

$$U_{1-2} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = \int_{s_1}^{s_2} F \cos\theta \, ds \qquad (4.1)$$

Then the area under this graph bounded by s_1 and s_2 represents the total work, Fig. 4.1b.

4.2 Work of a Constant Force Moving Along a Straight Line

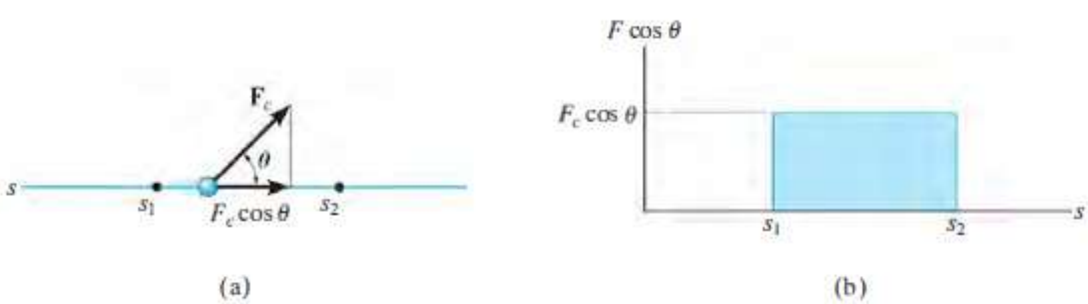


Fig. 4.2

If the force $\boldsymbol{F_c}$ has a constant magnitude a onstant angle θ from its straight-line path, Fig. 4.2a, then the component of $\boldsymbol{F_c}$ in the direction of displacement is always $F_c \cos\theta$. The work done by $\boldsymbol{F_c}$ when the particle is displaced from s_1 to s_2 is determined from Eq. 4.1, in which case

$$U_{1-2} = F_c \cos\theta \int_{s_1}^{s_2} ds$$

or

$$U_{1-2} = F_c \cos\theta (s_2 - s_1) \qquad (4.2)$$

Here the work of $\boldsymbol{F_c}$ represents the area of the rectangle in Fig. 4.2b.

4.3 Work of a Weight

Thus, the work is independent of the path and is equal to the magnitude of the particle's weight times its vertical displacement.

$$U_{1-2} = -W\ \Delta y \qquad (4.3)$$

Note that the work is negative, if W is downward and Δy is upward (i.e +ve).

4.4 Work of a Spring Force

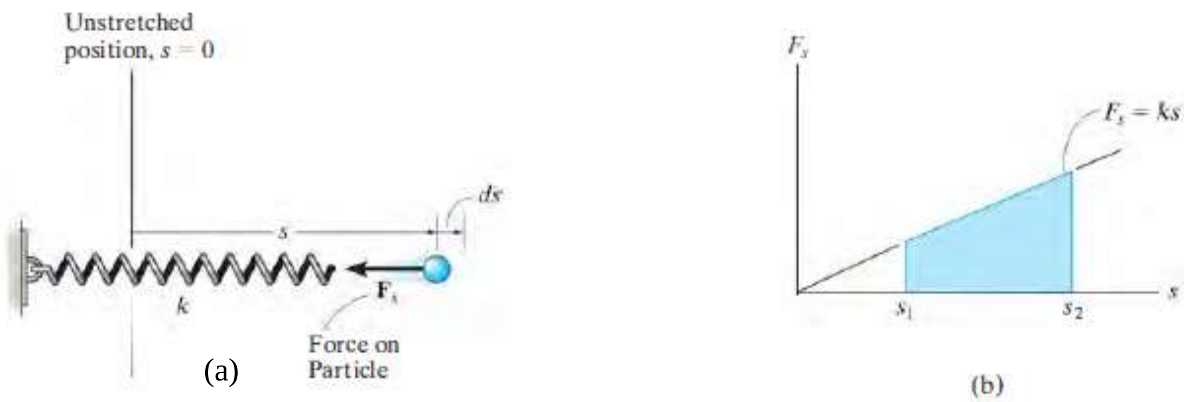


Fig. 4.3

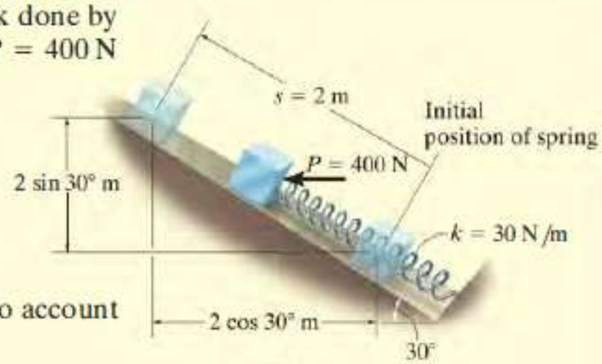
If an elastic spring is elongated a distance ds, Fig. 4.3a, then the work done by the force that acts on the attached particle is $dU = -F_s ds = -ksds$. The work is negative since $\boldsymbol{F_s}$ acts in the opposite sense to ds. If the particle displaces from s_1 to s_2, the work of $\boldsymbol{F_s}$ is then

$$U_{1-2} = -\int_{s_1}^{s_2} F_s ds = -\int_{s_1}^{s_2} ksds$$

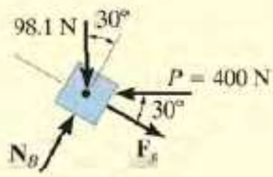
$$U_{1-2} = -\left(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2\right) \qquad (4.4)$$

Example 4.1

The 10-kg block shown in Fig. a rests on the smooth incline. If the spring is originally stretched 0.5 m, determine the total work done by all the forces acting on the block when a horizontal force $P = 400$ N pushes the block up the plane $s = 2$ m.



(a)



(b)

SOLUTION

First the free-body diagram of the block is drawn in order to account for all the forces that act on the block, Fig. b

Horizontal Force *P*. Since this force is *constant*, the work is determined using Eq. 4.2. The result can be calculated as the force times the component of displacement in the direction of the force; i.e.,

$$U_P = 400 \text{ N } (2 \text{ m} \cos 30°) = 692.8 \text{ J}$$

or the displacement times the component of force in the direction of displacement, i.e.,

$$U_P = 400 \text{ N} \cos 30°(2 \text{ m}) = 692.8 \text{ J}$$

Spring Force F_s. In the initial position the spring is stretched $s_1 = 0.5$ m and in the final position it is stretched $s_2 = 0.5 \text{ m} + 2 \text{ m} = 2.5$ m. We require the work to be negative since the force and displacement are opposite to each other. The work of $\mathbf{F}_s$ is thus

$$U_s = -\left[\tfrac{1}{2}(30 \text{ N/m})(2.5 \text{ m})^2 - \tfrac{1}{2}(30 \text{ N/m})(0.5 \text{ m})^2\right] = -90 \text{ J}$$

Weight *W*. Since the weight acts in the opposite sense to its vertical displacement, the work is negative; i.e.,

$$U_W = -(98.1 \text{ N})\,(2 \text{ m} \sin 30°) = -98.1 \text{ J}$$

Note that it is also possible to consider the component of weight in the direction of displacement; i.e.,

$$U_W = -(98.1 \sin 30° \text{ N})\,(2 \text{ m}) = -98.1 \text{ J}$$

Normal Force N_B. This force does *no work* since it is *always* perpendicular to the displacement.

Total Work. The work of all the forces when the block is displaced 2 m is therefore

$$U_T = 692.8 \text{ J} - 90 \text{ J} - 98.1 \text{ J} = 505 \text{ J} \qquad \textit{Ans.}$$

4.5 Principle of Work and Energy

$$\Sigma U_{1-2} = \tfrac{1}{2}mv_2^2 - \tfrac{1}{2}mv_1^2 \qquad (4.5)$$

This equation represents the principle of work and energy for the particle. The term on the left is the sum of the work done by all the forces acting on the particle as the particle moves from point 1 to point 2. The two terms on the right side, which are of the form $T = \frac{1}{2}mv^2$, define the particle's final and initial kinetic energy, respectively. Like work, kinetic energy is a scalar and has units of joules (J). However, unlike work, which can be either positive or negative, the kinetic energy is always positive, regardless of the direction of motion of the particle.

When Eq. 4.5 is applied, it is often expressed in the form

$$\boxed{T_1 + \Sigma U_{1-2} = T_2} \qquad (4.6)$$

which states that the particle's initial kinetic energy plus the work done by all the forces acting on the particle as it moves from its initial to its final position is equal to the particle's final kinetic energy.

4.6 Work of Friction Caused by Sliding

A special class of problems will now be investigated which requires a careful application of Eq. 4.6. These problems involve cases where a body slides over the surface of another body in the presence of friction.

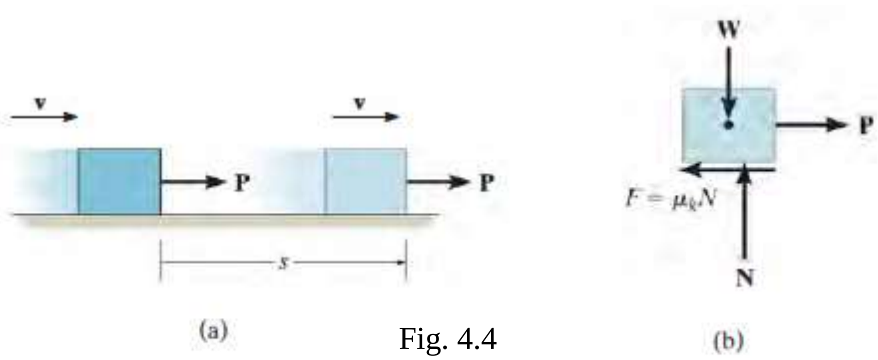


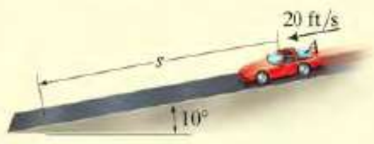
Fig. 4.4

If the applied force P just balances the resultant frictional force $\mu_k N$, Fig. 4.4b, then due to equilibrium a constant velocity v is maintained, and one would expect Eq. 4.6 to be applied as follows:

$$\tfrac{1}{2}mv^2 + Ps - \mu_k Ns = \tfrac{1}{2}mv^2 \qquad (4.7)$$

Indeed this equation is satisfied if $P = \mu_k N$; however, as one realizes from experience, the sliding motion will generate heat, a form of energy which seems not to be accounted for in the work-energy equation.

Example 4.2

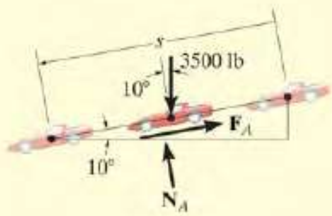


(a)

The 3500-lb automobile shown in Fig. a travels down the 10° inclined road at a speed of 20 ft/s. If the driver jams on the brakes, causing his wheels to lock, determine how far s the tires skid on the road. The coefficient of kinetic friction between the wheels and the road is $\mu_k = 0.5$.

SOLUTION

This problem can be solved using the principle of work and energy, since it involves force, velocity, and displacement.



(b)

Work (Free-Body Diagram). As shown in Fig. b, the normal force $\mathbf{N}_A$ does no work since it never undergoes displacement along its line of action. The weight, 3500 lb, is displaced $s \sin 10°$ and does positive work. Why? The frictional force $\mathbf{F}_A$ does both external and internal work when it undergoes a displacement s. This work is negative since it is in the opposite sense of direction to the displacement. Applying the equation of equilibrium normal to the road, we have

$$+\nwarrow \Sigma F_n = 0; \quad N_A - 3500 \cos 10° \text{ lb} = 0 \qquad N_A = 3446.8 \text{ lb}$$

Thus,

$$F_A = \mu_k N_A = 0.5\,(3446.8 \text{ lb}) = 1723.4 \text{ lb}$$

Principle of Work and Energy.

$$T_1 + \Sigma U_{1\text{-}2} = T_2$$

$$\frac{1}{2}\left(\frac{3500 \text{ lb}}{32.2 \text{ ft/s}^2}\right)(20 \text{ ft/s})^2 + 3500 \text{ lb}(s \sin 10°) - (1723.4 \text{ lb})s = 0$$

Solving for s yields

$$s = 19.5 \text{ ft} \qquad \textit{Ans.}$$

NOTE: If this problem is solved by using the equation of motion, *two steps* are involved. First, from the free-body diagram, Fig. b, the equation of motion is applied along the incline. This yields

$$+\swarrow \Sigma F_s = ma_s; \quad 3500 \sin 10° \text{ lb} - 1723.4 \text{ lb} = \frac{3500 \text{ lb}}{32.2 \text{ ft/s}^2}a$$

$$a = -10.3 \text{ ft/s}^2$$

Then, since a is constant, we have

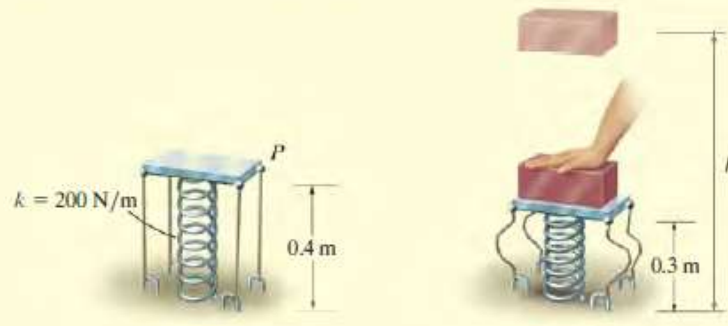
$$(+\swarrow) \quad v^2 = v_0^2 + 2a_c(s - s_0);$$

$$(0)^2 = (20 \text{ ft/s})^2 + 2(-10.3 \text{ ft/s}^2)(s - 0)$$

$$s = 19.5 \text{ ft} \qquad \textit{Ans.}$$

Example 4.3

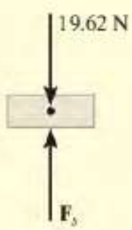
The platform P, shown in Fig. a, has negligible mass and is tied down so that the 0.4-m-long cords keep a 1-m-long spring compressed 0.6 m when *nothing* is on the platform. If a 2-kg block is placed on the platform and released from rest after the platform is pushed down 0.1 m, Fig. b, determine the maximum height h the block rises in the air, measured from the ground.



(a) (b)

SOLUTION

Work (Free-Body Diagram). Since the block is released from rest and later reaches its maximum height, the initial and final velocities are zero. The free-body diagram of the block when it is still in contact with the platform is shown in Fig. c. Note that the weight does negative work and the spring force does positive work. Why? In particular, the *initial compression* in the spring is $s_1 = 0.6\text{ m} + 0.1\text{ m} = 0.7\text{ m}$. Due to the cords, the spring's *final compression* is $s_2 = 0.6\text{ m}$ (after the block leaves the platform). The bottom of the block rises from a height of $(0.4\text{ m} - 0.1\text{ m}) = 0.3\text{ m}$ to a final height h.



(c)

Principle of Work and Energy.

$$T_1 + \Sigma U_{1\text{-}2} = T_2$$

$$\tfrac{1}{2}mv_1^2 + \left\{-\left(\tfrac{1}{2}ks_2^2 - \tfrac{1}{2}ks_1^2\right) - W\,\Delta y\right\} = \tfrac{1}{2}mv_2^2$$

Note that here $s_1 = 0.7\text{ m} > s_2 = 0.6\text{ m}$ and so the work of the spring as determined from Eq. 14–4 will indeed be positive once the calculation is made. Thus,

$$0 + \left\{-\left[\tfrac{1}{2}(200\text{ N/m})(0.6\text{ m})^2 - \tfrac{1}{2}(200\text{ N/m})(0.7\text{ m})^2\right] - (19.62\text{ N})[h - (0.3\text{ m})]\right\} = 0$$

Solving yields

$$h = 0.963\text{ m} \qquad \textit{Ans.}$$

Problem Set 4.1

1. The 3500 N automobile shown in Example 4.2 travels down the 10° inclined road at a speed of 6 m/s. If the driver jams on the brakes, causing his wheels to lock, determine how far s the tires skid on the road. The coefficient of kinetic friction between the wheels and the road is $\mu_k = 0.5$.

2. Apply the principles of Work and Energy to determine the total work done on the bodies/objects in Chapter 3.